

Inverse of multivector: Beyond $p+q=5$ threshold

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- 1 Introduction
- 2 Grade-negated self-product
- 3 Inverse of general MV of algebras of dimension $n = p + q \leq 5$
- 4 Inverse of general MV of algebras of dimension $n = p + q = 6$
- 5 Conclusions and perspective

Problem formulation and notations

- General Clifford algebra multivector (MV)
 $A = \text{scalar} + \text{bivector} + \dots + \text{pseudoscalar}$ will be considered in this report
- The problem is to find inverse of general MV A , i. e. A^{-1} for which $AA^{-1} = A^{-1}A = 1$ holds,
- $n = p + q$ in the title indicates that formula for A^{-1} depends only on dimension of the Clifford algebra $Cl_{p,q}$ and not on particular signature (p, q)
- The search of solution of the problem heavily relies on computer-aided symbolic mathematics

So before explaining calculation of inverse of MV a brief outlook of the *Mathematica* package that was written for this purpose will be presented.

GeometricAlgebra package for *Mathematica*

Main features

- Textbook notation (implemented without *Mathematica* palettes) and automatic precedence of inner, outer and geometric products
- Algebraic operations in orthonormal frame (additive representation, symbolic coefficients)
- Switching between multiple algebras (denoted by different color) in same *Mathematica* session
- Matrix representations of $Cl_{p,q}$ algebras
- Idempotents with different base element sorting
- Main involutions and general multivector inverse

Files

- GAC.Base.nb (main file with installation information)
- UnicodeCharactersAdd.tr (add-on for *Mathematica* system file)
- Start.nb (first usage example with short introduction)
- AlgebraFormulary.nb (Clifford algebra formulary)
- ElementaryFunctions.nb (Clifford function expansions)
- Idempotents.nb (algebras idempotents)
- Blades (general blade formulas test)
- MultivectorInverse (general multivector inverses and formula test)
- MatrixRepresentations.nb (Clifford algebra matrix representations)

May be download under licence
GPL-3.0 from <https://github.com/ArturasAcus/GeometricAlgebra>

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Grade-negated self-product

Grade- r negation operation

- Changes the sign of grade- r part of the multivector A
- It is an **involution** denoted as $A_{\bar{r}}$. Note the bar over negated grade index.
- Formal definition

$$A_{\bar{r}} = A - 2\langle A \rangle_r$$
- Main properties:
 $(A_{\bar{p}})_{\bar{r}} = A_{\bar{p},\bar{r}} = A_{\bar{r},\bar{p}}, A_{\bar{r},\bar{r}} = A,$
 $(A + B)_{\bar{r}} = A_{\bar{r}} + B_{\bar{r}}$
- However, $(AB)_{\bar{r}} \neq A_{\bar{r}}B_{\bar{r}}$
- Commutator

$$AA_{\bar{r}} - A_{\bar{r}}A = 2(\langle A \rangle_r A - A \langle A \rangle_r)$$

Relation to standard involutions

- Reversion $\tilde{A} = A_{\bar{2},\bar{3},\bar{6},\bar{7},\bar{10},\bar{11}}\dots$
- Grade inversion
 $\hat{A} = A_{\bar{1},\bar{3},\bar{5},\bar{7},\bar{9},\bar{11}}\dots$
- Clifford conjugate
 $\hat{\hat{A}} = A_{\bar{1},\bar{2},\bar{5},\bar{6},\bar{9},\bar{10}}\dots$

Important: grades 0; 4; 8; 12; $\dots$ are absent in the above list of standard GA involutions. They play special role in inverse formulas for $n > 5$.

Grade-negated self-product

Definitions and the idea

Grade-negated self-product is defined via grade- r negation operation.

- Grade-negated self-product of general multivector A is geometric product of A with any number of its grade-negated counterparts. Examples $\color{red}{A}A_{\bar{k},\bar{l},\bar{t}}A_{\bar{s},\bar{t}}\cdots$ or $A_{\bar{k},\bar{l},\bar{t}}A_{\bar{s},\bar{t}}\cdots\color{red}{A}$. operation.
- Self-products where the initial MV A stands in the left-most or in the right-most position are of special importance in finding the inverse multivector A^{-1} .
- Any identity, which contains only operations of geometric product and grade negations, and which was derived in particular Clifford algebra of dimension $n = p + q$ will also be true in any n -dimensional algebra of arbitrary signature (p, q) .
- Assume we can construct a **scalar** s_m using just grade-negated self-products of m multivectors with the initial multivector A standing either in the left-most or right-most position.

Grade-negated self-product

Definitions and the idea (continued)

- Name this way obtained scalar the **discriminant** of A.

scalar : $s_m = \overbrace{AA_{\bar{r}, \bar{t}, \dots} A_{\bar{s}, \dots}}^m \dots$

- If we succeed in constructing such a scalar then left/right inverse of MV can be obtained by removing left or right most positioned initial MV A from the scalar $s_m = AA_{\bar{r}, \bar{t}, \dots} A_{\bar{s}, \dots} \dots$ and then dividing the result by s_m :

$$A^{-1} = \frac{A_{\bar{r}, \bar{t}, \dots} A_{\bar{s}, \dots} \dots}{s_m}.$$

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Inverses in dimensions $n = p + q \leq 5$

Two dimensional algebras: $Cl_{2,0}$ case

- Write $A = \sum_{J=0}^{2^n-1} a_J \mathbf{e}_J$, where the multi-index J covers all orthonormal base elements arranged in inverse degree lexicographic order, i. e., the lowest grade elements appear first while elements of the same grade are ordered lexicographically:
 $J = [\{\}, \{1\}, \{2\}, \{1, 2\}]$.
- Check that self-product $AA_{\bar{1},\bar{2}}$ immediately gives the required scalar $s_2 = a_{\{\}}^2 - a_1^2 - a_2^2 + a_{1,2}^2$.

Coefficient notation

- It is more convenient to rewrite the coefficient indices as $s_2 = a_0^2 - a_1^2 - a_2^2 + a_3^2$, i. e. the index of scalar a_0 is zero, the coefficients at vectors run from 1 to n while numerical indices of higher grades increase monotonically up to 2^n .
- General rule for grade- r index: start the element index by $\sum_{k=0}^{r-1} \binom{n}{k}$ and end by $\left(\sum_{k=0}^{r-1} \binom{n}{k}\right) + \binom{n}{r} - 1$. Here $\binom{n}{k}$ is the binomial coefficient.

Inverses in dimensions $n = p + q \leq 5$

Notes about discriminant

- The discriminant s_m comprises all coefficients a_i of multivector A
- The MV has inverse if discriminant is unequal to zero, $s_m \neq 0$. This imposes the requirement for MV coefficients.
- Because reverse leaves the discriminant invariant, the formula for $n = 2$ can be rewritten as $s_2 = AA_{\bar{1},\bar{2}} = A_{\bar{1},\bar{2}}A$, from which the left/right inverse formulas follow,

$$A^{-1} = A_{\bar{1},\bar{2}}/(AA_{\bar{1},\bar{2}}); \quad A^{-1} = A_{\bar{1},\bar{2}}/(A_{\bar{1},\bar{2}}A).$$

- The eliminated grades are grade-negated in one of MVs in the self-product. In the example the grades 1 and 2 have been eliminated simultaneously by $A_{\bar{1},\bar{2}}$. This rule is valid for discriminants up to $n \leq 5$.
- In all cases the highest degree of coefficients in the discriminant always matches the degree of determinant of real matrix representation of A , in this case 2×2 real matrices (next slide)

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- In all cases the highest degree of coefficients in the discriminant always matches the degree of determinant of real matrix representation of A , in this case 2×2 real matrices (**next slide**)

Inverses in dimensions $n = p + q \leq 5$

Period-8 table of irreducible matrix representations of GA for $Cl_{p,q}$, where p and q indicate positive and negative signature.

$p \backslash q$	0	1	2	3	4	5	6	7
0	$\mathbb{R}$	$\mathbb{C}$	$\mathbb{H}$	${}^2\mathbb{H}$	$\mathbb{H}(2)$	$\mathbb{C}(4)$	$\mathbb{R}(8)$	${}^2\mathbb{R}(8)$
1	${}^2\mathbb{R}$	$\mathbb{R}(2)$	$\mathbb{C}(2)$	$\mathbb{H}(2)$	${}^2\mathbb{H}(2)$	$\mathbb{H}(4)$	$\mathbb{C}(8)$	$\mathbb{R}(16)$
2	$\mathbb{R}(2)$	${}^2\mathbb{R}(2)$	$\mathbb{R}(4)$	$\mathbb{C}(4)$	$\mathbb{H}(4)$	${}^2\mathbb{H}(4)$	$\mathbb{H}(8)$	$\mathbb{C}(16)$
3	$\mathbb{C}(2)$	$\mathbb{R}(4)$	${}^2\mathbb{R}(4)$	$\mathbb{R}(8)$	$\mathbb{C}(8)$	$\mathbb{H}(8)$	${}^2\mathbb{H}(8)$	$\mathbb{H}(16)$
4	$\mathbb{H}(2)$	$\mathbb{C}(4)$	$\mathbb{R}(8)$	${}^2\mathbb{R}(8)$	$\mathbb{R}(16)$	$\mathbb{C}(16)$	$\mathbb{H}(16)$	${}^2\mathbb{H}(16)$
5	${}^2\mathbb{H}(2)$	$\mathbb{H}(4)$	$\mathbb{C}(8)$	$\mathbb{R}(16)$	${}^2\mathbb{R}(16)$	$\mathbb{R}(32)$	$\mathbb{C}(32)$	$\mathbb{H}(32)$
6	$\mathbb{H}(4)$	${}^2\mathbb{H}(4)$	$\mathbb{H}(8)$	$\mathbb{C}(16)$	$\mathbb{R}(32)$	${}^2\mathbb{R}(32)$	$\mathbb{R}(64)$	$\mathbb{C}(64)$
7	$\mathbb{C}(8)$	$\mathbb{H}(8)$	${}^2\mathbb{H}(8)$	$\mathbb{H}(16)$	$\mathbb{C}(32)$	$\mathbb{R}(64)$	${}^2\mathbb{R}(64)$	$\mathbb{R}(128)$

The left superscript denotes direct sum of matrices, ${}^2\mathbb{R}(2) \equiv \mathbb{R}(2) \oplus \mathbb{R}(2)$.

Inverses in dimensions $n = p + q \leq 5$

Three dimensional algebras: $Cl_{2,1}$ case

- Our goal is to eliminate as many grades as possible by forming suitable self-products, for example, for $Cl_{2,1}$ algebra.
- The $A = a_0 + a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + a_3\mathbf{e}_3 + a_4\mathbf{e}_{12} + a_5\mathbf{e}_{13} + a_6\mathbf{e}_{23} + a_7\mathbf{e}_{123}$ product with $A_{\bar{1},\bar{2}}$ eliminates grades 1 and 2. The result is the new multivector $B = AA_{\bar{1},\bar{2}} = b_0 + b_7\mathbf{e}_{123}$ which consists of scalar and grade-3 element. In particular, $b_0 = a_0^2 - a_1^2 - a_2^2 + a_3^2 + a_4^2 - a_5^2 - a_6^2 + a_7^2$ and $b_7 = -2a_3a_4 + 2a_2a_5 - 2a_1a_6 + 2a_0a_7$.
- Repeating the grade negation procedure for grade-3 then yields left-handed discriminant

$$s_4 = BB_{\bar{3}} = AA_{\bar{1},\bar{2}}(AA_{\bar{1},\bar{2}})_{\bar{3}} = b_0^2 - b_7^2.$$

- Reversion of s_4 immediately gives the A-right-handed discriminant

$$s'_4 = (A_{\bar{1},\bar{2}}A)_{\bar{3}}A_{\bar{1},\bar{2}}A$$

that contains four MVs (**next slide**).

Inverses in dimensions $n = p + q \leq 5$

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Inverses in dimensions $n = p + q \leq 5$

Three dimensional algebras: $Cl_{2,1}$ case (continued)

- For other 3D algebras, namely $Cl_{3,0}$, $Cl_{1,2}$, and $Cl_{0,3}$ the same formula gives different fully expanded expressions, with other signs of individual coefficients a_i .
- The discriminant is determined by squares of two coefficients, b_0^2 and b_7^2 , the difference of which should not be zero for invertible multivector A to exist.

Inverses in dimensions $n = p + q \leq 5$

Four dimensional algebras: $n = p + q = 4$

- In this case elimination of grades can proceed in two alternative ways.
- 1) Negate simultaneously grades 1 and 2 and then in the next step the grades 3 and 4, i. e. $s_4 = \mathbf{A} \mathbf{A}_{\bar{1}, \bar{2}} (\mathbf{A} \mathbf{A}_{\bar{1}, \bar{2}})_{\bar{3}, \bar{4}}$
- 2) Eliminate 2 and 3, and then 1 and 4 grades, i. e.
 $s_4 = \mathbf{A} \mathbf{A}_{\bar{2}, \bar{3}} (\mathbf{A} \mathbf{A}_{\bar{2}, \bar{3}})_{\bar{1}, \bar{4}}$.
- Similar elimination can be done in right-hand versions,
 $s'_4 = (\mathbf{A}_{\bar{2}, \bar{3}} \mathbf{A})_{\bar{1}, \bar{4}} \mathbf{A}_{\bar{2}, \bar{3}} \mathbf{A} = (\mathbf{A}_{\bar{1}, \bar{2}} \mathbf{A})_{\bar{3}, \bar{4}} \mathbf{A}_{\bar{1}, \bar{2}} \mathbf{A}$.
- Despite different forms, expanded explicit expressions for discriminants s_m and s'_m in fact always appear to be equal, $s_m = s'_m$.
- The appearance of symbol $\mathbf{A}$ as much as four times in the products s_4 and s'_4 is again what we expect from the matrix representation for $n = 4$ algebras, (next slide) namely, the highest degree of determinant polynomial should be of degree 4 as well, in accord with 4×4 matrix representation $\mathbb{R}(4)$.

Inverses in dimensions $n = p + q \leq 5$

Period-8 table of irreducible matrix representations of GA for $Cl_{p,q}$, where p and q indicate positive and negative signature.

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Inverses in dimensions $n = p + q \leq 5$

Five dimensional algebras: $n = p + q = 5$

- 1) Eliminate simultaneously grades 2 and 3
- 2) Then eliminate grades 1 and 4
- 3) Finally eliminate grade 5. Apart from the last step this sequence is exactly the same as the second alternative of $n = 4$ case.
- The final result is

$$s'_8 = (FA)_{\bar{5}} F \mathbf{A} \quad \text{with} \quad F = (A_{\bar{2},\bar{3}} A)_{\bar{1},\bar{4}} A_{\bar{2},\bar{3}}$$

for right-hand sided form of the discriminant and

$$s_8 = \mathbf{A} D (AD)_{\bar{5}} \quad \text{with} \quad D = A_{\bar{2},\bar{3}} (A A_{\bar{2},\bar{3}})_{\bar{1},\bar{4}}$$

for left-hand discriminant.

- Matrix representation of five dimensional Clifford algebras corresponds to real ${}^2\mathbb{R}(4)$ matrix (**next slide**) the the determinant of which is a polynomial of degree 8. This is in accord with the structure of discriminant $s_8 = s'_8$ where the A occurs as much as eight times.

Inverses in dimensions $n = p + q \leq 5$

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Inverses in dimensions $n = p + q \leq 5$

The case of even multivectors

- Even multivectors make subalgebra
- There is an isomorphisms

$$Cl_{p,q+1}^+ \cong Cl_{p,q}, \quad Cl_{p+1,q}^+ \cong Cl_{q,p}$$

- This can help to get inverse for MV of lower dimensional algebra once formula of higher dimension algebra is known.
- Example for $p + q = 4$. Formula of inverse of general MV for $n = 4$ is

$$A^{-1} = \frac{A_{\bar{2},\bar{3}}(AA_{\bar{2},\bar{3}})_{\bar{1},\bar{4}}}{AA_{\bar{2},\bar{3}}(AA_{\bar{2},\bar{3}})_{\bar{1},\bar{4}}}$$

- If A consists only of even grades then A is invariant to odd grade negations, so that the above formula for even MV's reduces to

$$A^{-1} = \frac{A_{\bar{2}}(AA_{\bar{2}})_{\bar{4}}}{AA_{\bar{2}}(AA_{\bar{2}})_{\bar{4}}}$$

Inverses in dimensions $n = p + q \leq 5$

The case of even multivectors (continued)

- Remember that even grade elements of $Cl_{1,3}$ go to vectors and bivectors of $Cl_{3,0}$ and the pseudoscalar I_4 goes to pseudoscalar I_3 . Thus, in $Cl_{3,0}$ algebra notation the last formula should be rewritten as

$$A^{-1} = \frac{A_{\bar{1},\bar{2}}(AA_{\bar{1},\bar{2}})_{\bar{3}}}{AA_{\bar{1},\bar{2}}(AA_{\bar{1},\bar{2}})_{\bar{3}}}$$

which coincides with the result for $p + q = 3$

- An alternative formulas for even multivectors exist, which includes even MV multiplication by vector (**next slide**)
- The inverse of **even MV** for $p + q = 6$ can be written as a self-negated product

Inverses in dimensions $n = p + q \leq 5$

The case of even multivectors (continued)

$Cl_{p,q}$	$(A^+)^{-1}$	
$p + q = 2$	$\frac{B(A^+B)}{(A^+B)^2},$	$B = (e_i)_{\bar{1}} = -e_i$
$p + q = 3$	$\frac{C(A^+C)}{(A^+C)^2},$	$C = (A^+e_i)_{\bar{1}}$
$p + q = 4$	$\frac{C(A^+C)}{(A^+C)^2},$	$C = (A^+e_i)_{\bar{1}}$
$p + q = 5$	$\frac{D(A^+D)}{(A^+D)^2},$	$D = (A^+e_i)_{\bar{3}}(A^+(A^+e_i)_{\bar{3}})_{\bar{1}}$
$p + q = 6$	$\frac{D(A^+D)_{\bar{6}}}{(A^+D)(A^+D)_{\bar{6}}},$	$D = (A^+e_i)_{\bar{3}}(A^+(A^+e_i)_{\bar{3}})_{\bar{1}}$

Inverses in dimensions $n = p + q \leq 5$

The case of even multivectors (continued)

The denominators in inverse formulas are the discriminants of respective multivectors

$$s_2 = (A+B)^2/\mathbf{e}_i^2, \quad s'_4 = (A+C)^2/\mathbf{e}_i^2, \quad s_4 = (A+D)^2/\mathbf{e}_i^4$$

and

$$s_8 = (A+D)(A+D)_{\bar{6}}/\mathbf{e}_i^4.$$

Here explicit $\mathbf{e}_i^2 = \pm 1$ and $\mathbf{e}_i^4 = +1$ factors take into account the correct sign and norm once the orthonormal base vector $\mathbf{e}_i$ is replaced by arbitrary unnormalized and nonisotropic vector v .

Can grade elimination procedure be extended beyond $n = 5$?

The short answer is 'yes' if we allow **multilinear combinations** of grade-negated self-products with properly chosen coefficients.

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Inverse in dimension $n = p + q = 6$

Why single self-negated product does not work in $p + q = 6$?

- The culprit appear to be the **grade-4** blades. They cannot be eliminated by single self-product!
- For $n = 6$ we can simultaneously eliminate either grades 1, 2, 5 and 6 (then 0, 3 and 4 grades survive) or, alternatively, the grades 2, 3 and 6 (then grades 0, 1, 4 and 5 persist).
- There exists a subalgebra with base formed by elements $\{1, \mathbf{e}_{1256}, \mathbf{e}_{1346}, \mathbf{e}_{2345}\}$ such that any self product of multivector

$$B = a_0 + a_{47}\mathbf{e}_{1256} + a_{49}\mathbf{e}_{1346} + a_{52}\mathbf{e}_{2345}$$

by grade-negated B will yield new MV having at least one grade-4 base element present.

- There is **only one** such an "uneliminatable" grade part in $p + q = 6$ case.

We will start by trying to eliminate this special grade-4 part.

Inverse in dimension $n = p + q = 6$

How many terms we need in product?

$p \backslash q$	0	1	2	3	4	5	6	7
0	$\mathbb{R}$	$\mathbb{C}$	$\mathbb{H}$	${}^2\mathbb{H}$	$\mathbb{H}(2)$	$\mathbb{C}(4)$	$\mathbb{R}(8)$	${}^2\mathbb{R}(8)$
1	${}^2\mathbb{R}$	$\mathbb{R}(2)$	$\mathbb{C}(2)$	$\mathbb{H}(2)$	${}^2\mathbb{H}(2)$	$\mathbb{H}(4)$	$\mathbb{C}(8)$	$\mathbb{R}(16)$
2	$\mathbb{R}(2)$	${}^2\mathbb{R}(2)$	$\mathbb{R}(4)$	$\mathbb{C}(4)$	$\mathbb{H}(4)$	${}^2\mathbb{H}(4)$	$\mathbb{H}(8)$	$\mathbb{C}(16)$
3	$\mathbb{C}(2)$	$\mathbb{R}(4)$	${}^2\mathbb{R}(4)$	$\mathbb{R}(8)$	$\mathbb{C}(8)$	$\mathbb{H}(8)$	${}^2\mathbb{H}(8)$	$\mathbb{H}(16)$
4	$\mathbb{H}(2)$	$\mathbb{C}(4)$	$\mathbb{R}(8)$	${}^2\mathbb{R}(8)$	$\mathbb{R}(16)$	$\mathbb{C}(16)$	$\mathbb{H}(16)$	${}^2\mathbb{H}(16)$
5	${}^2\mathbb{H}(2)$	$\mathbb{H}(4)$	$\mathbb{C}(8)$	$\mathbb{R}(16)$	${}^2\mathbb{R}(16)$	$\mathbb{R}(32)$	$\mathbb{C}(32)$	$\mathbb{H}(32)$
6	$\mathbb{H}(4)$	${}^2\mathbb{H}(4)$	$\mathbb{H}(8)$	$\mathbb{C}(16)$	$\mathbb{R}(32)$	${}^2\mathbb{R}(32)$	$\mathbb{R}(64)$	$\mathbb{C}(64)$
7	$\mathbb{C}(8)$	$\mathbb{H}(8)$	${}^2\mathbb{H}(8)$	$\mathbb{H}(16)$	$\mathbb{C}(32)$	$\mathbb{R}(64)$	${}^2\mathbb{R}(64)$	$\mathbb{R}(128)$

We need 8 term product. Too many. Fortunately the determinant of matrix representation of scalar+grade-4 multivector is a **square!**, i. e. $\det(\langle A \rangle_4) = (\dots)^2$. Only 4 term product multilinear combination should eliminate grade-4 part.

Inverse in dimension $n = p + q = 6$

Pattern of multilinear combination

- Form prototype of a geometric product of the form

$$s_{4+4} = s_{4f} + s_{4g} =$$

$$b_1 B f_5 \left(f_4(B) f_3 \left(f_2(B) f_1(B) \right) \right) + b_2 B g_5 \left(g_4(B) g_3 \left(g_2(B) g_1(B) \right) \right),$$

where f_j, g_j denotes yet undetermined grade-4 negations and b_1, b_2 are unknown scalar coefficients of the multilinear combination.

- Not unique. The other three possible choices, however, yield the same answer.

Calculate explicit symbolic form of square root of the determinant of matrix representation of the multivector

$B = a_0 + a_{47}\mathbf{e}_{1256} + a_{49}\mathbf{e}_{1346} + a_{52}\mathbf{e}_{2345}$ and compare it with all possible negations of grade-4 substituted for f_j and g_j .

Inverse in dimension $n = p + q = 6$

Modelling the grade negation

- Multiply the grade-4 part by unknown coefficients p_{4jk} , where the index j denotes the involution number f_j, g_j in the self-product and k is the term in multilinear combination (in $p + q = 6$ case $k = 1$ for f and $k = 2$ for g).
- The coefficients p_{ijk} acquire the values ± 1 only, where -1 means that involution that changes sign of grade i is applied and $+1$ means that grade- i remains unchanged.
- The answer: $\{b_1 = -2/3, b_2 = -1/3, p_{411} = -1, p_{412} = 1, p_{421} = -1, p_{422} = 1, p_{431} = -1, p_{432} = -1, p_{441} = -1, p_{442} = 1, p_{451} = -1, p_{452} = 1\}$ and $\{b_1 = -1/3, b_2 = -2/3, p_{411} = 1, p_{412} = -1, p_{421} = 1, p_{422} = -1, p_{431} = -1, p_{432} = -1, p_{441} = 1, p_{442} = -1, p_{451} = 1, p_{452} = -1\}$.
- Check the answer with the most general **scalar + grade-4** multivector.

Inverse in dimension $n = p + q = 6$

The full discriminant

- Repeat the procedure for involutions that consecutively eliminate all other grades. How to double the number of multivectors?
- Simultaneously either grades 1, 2, 5 and 6 (then grades 0, 3 and 4 remain), or, alternatively grades 2, 3 and 6 (then grades 0, 1, 4 and 5 remain) can be simultaneously eliminated. All in all there are $\frac{6!}{2!4!} + \frac{6!}{3!3!} + 1 = 36$ elements of grades 2, 3 and 6 which can be eliminated simultaneously. This is more than 28 total elements of grades 1, 2, 5 and 6 that need to be eliminated.
- Therefore, if we replace B in s_{4+4} pattern by self-negated product $AA_{\bar{2},\bar{3},\bar{6}}$ we will have eight required terms.
- Then we find the other involutions using the multivector $a_0 + a_1\mathbf{e}_1 + a_{22}\mathbf{e}_{123} + a_{47}\mathbf{e}_{1256}$ and check result with general MV.

30 valid meaningfull solutions, which fall into two classes. Among them there is two solution with minimal number of negated MVs equal to 34.

Inverse in dimension $n = p + q = 6$

The answer

 $Cl_{p,q}$ A^{-1}

$p + q = 0$

$\frac{B}{AB},$

$B = 1$

$p + q = 1$

$\frac{B(AB)_{\bar{1}}}{AB(AB)_{\bar{1}}} = \frac{\widehat{A}}{AA},$

$B = 1$

$p + q = 2$

$\frac{C}{AC} = \frac{\widehat{\widehat{A}}}{AA},$

$C = A_{\bar{1},\bar{2}}$

$p + q = 3$

$\frac{C(AC)_{\bar{3}}}{AC(AC)_{\bar{3}}} = \frac{\widehat{\widehat{\widehat{A}}}}{AAAA},$

$C = A_{\bar{1},\bar{2}}$

$p + q = 4$

$\frac{D}{AD},$

$D = A_{\bar{2},\bar{3}}(AA_{\bar{2},\bar{3}})_{\bar{1},\bar{4}}$

or $D = A_{\bar{1},\bar{2}}(AA_{\bar{1},\bar{2}})_{\bar{3},\bar{4}}$

$p + q = 5$

$\frac{D(AD)_{\bar{5}}}{AD(AD)_{\bar{5}}},$

$D = A_{\bar{2},\bar{3}}(AA_{\bar{2},\bar{3}})_{\bar{1},\bar{4}}$

$p + q = 6$

$\frac{G}{AG}, \quad G = \frac{1}{3}A_{\bar{2},\bar{3},\bar{6}}\left(H(HH)_{\bar{1},\bar{4},\bar{5}} + 2(H_{\bar{4}}(H_{\bar{4}}H_{\bar{4}})_{\bar{1},\bar{4},\bar{5}})_{\bar{4}}\right)$

or $G = \frac{1}{3}A_{\bar{2},\bar{3},\bar{6}}\left(\left(H(HH_{\bar{1},\bar{5}})_{\bar{4}}\right)_{\bar{1},\bar{5}} + 2(H_{\bar{4},\bar{5}}(H_{\bar{4},\bar{5}}H_{\bar{1},\bar{4}})_{\bar{4}})_{\bar{1},\bar{4}}\right)$

with $H = AA_{\bar{2},\bar{3},\bar{6}} = A\tilde{A}$

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Conclusions

- **Conjecture.** The inverse of a general MV of arbitrary Clifford algebra can be always expressed as a multilinear combination of proper number of specially constructed self-negated products, where in the discriminant expression the initial MV stays in the left-most or right-most position.
- The number of the multivectors in the negated products is determined by dimension of matrix representation of the algebra.
- **Hypothesis.** The minimal number of terms in the multilinear combination is related to the number of "uneliminatable" grades in the self product. There are no such grades in algebras with $p + q \leq 3$. There is only one grade, namely the grade-4, in algebras of dimension 4, 5, 6 and 7. The dimensions 4 and 5 are exceptional (not enough indices to form cyclic subgroup of the form scalar + grade-4). The algebras with $p + q = 8$ and $p + q = 9$ have two such special grades, 4 and 8.
- Computational performance? Multiplication of 8 general multivectors requires $(2^6)^8 = 281474976710656$ geometric multiplications of base elements in 6-dimensional algebra.

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A. Acus and A. Dargys,

submitted to Advances in Applied Clifford algebras

Thank You